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A Micro project on

**PREDICTIVE MAINTENANCE
USING MACHINE LEARNING**

Submitted in partial fulfillment of the

MACHINE LEARNING LAB

GRIET Lab On Board (G-LOB)

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CERTIFICATE

This is to certify that the micro project titled “**Predictive Maintenance Using Machine Learning**” is a bonafide work done by **P.Rakesh (23241A6752), Yashwanth(23241A6751) , Nithin Tej(23241A6745)** under **Machine Learning Lab - GRIET Lab On Board (G-LOB)** practice of our institute and that this work has not been submitted for the award of any other Degree/Diploma of any Institution/University.

Project Guide

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INTRODUCTION

1.1 Need for the project

The **Trending YouTube Video Statistics** project is needed to understand how videos gain popularity and what factors influence their rise on YouTube's trending list. With millions of videos uploaded daily, analyzing trends based on views, likes, comments, tags, and other engagement metrics helps uncover patterns behind viral content. This knowledge can benefit creators, marketers, and analysts by providing insights into audience behavior and the types of content that perform best across different countries.

Additionally, this project allows for deeper exploration of **cultural and regional differences** in video trends, showing how interests vary across nations like the US, India, and Japan. It also supports applications such as predicting future viral videos, improving content recommendation systems, and studying online engagement patterns over time. Overall, this analysis helps understand the dynamics of digital popularity and the algorithms that shape online visibility.

1.2 Project Description

The **Trending YouTube Video Statistics** project focuses on analyzing daily data of trending videos across multiple countries such as the USA, India, Japan, Germany, and more. The dataset includes important attributes like video title, channel name, publish time, views, likes, dislikes, comment count, and tags. By examining these features, the project aims to identify patterns and factors that influence a video's popularity and help understand what drives engagement on YouTube.

Through data cleaning, visualization, and statistical analysis, the project provides meaningful insights into viewer behavior, regional content preferences, and trending patterns over time. It can also be used to develop machine learning models that predict which types of videos are likely to trend in the future. Overall, the project serves as a valuable tool for content creators, marketers, and researchers to better understand and utilize YouTube's trending dynamics.

1.3 Components of the project

1. Data Collection

The dataset is collected using the **YouTube API**, which provides daily records of trending videos from multiple regions such as the US, IN, GB, DE, and FR.

Each file contains video details like title, channel, publish time, category, views, likes, dislikes, comment count, and tags.

2. Data Preprocessing

Cleaning the dataset by handling missing values, removing duplicates, and formatting dates and times.

Converting data types and merging datasets with category mapping JSON files for better understanding.

3. Exploratory Data Analysis (EDA)

Using visualizations to analyze trends in views, likes, dislikes, and comments.

Identifying the most popular categories, channels, and countries with the highest engagement.

Analyzing relationships between features such as likes vs views or comments vs engagement.

4. Statistical and Trend Analysis

Studying daily, weekly, and regional patterns of trending videos.

Comparing viewer behavior and content preferences across countries.

Finding factors that influence a video's trending status.

5. Report and Conclusion

Summarizing key findings, insights, and recommendations for creators or marketers.

Presenting how engagement metrics and video attributes affect popularity on YouTube

REQUIREMENT ANALYSIS

Software Requirements

- **Programming Language:**

Python 3.x is used for data cleaning, analysis, and visualization because of its simplicity and strong data science library support.

- **Libraries and Packages:**

- **pandas, numpy** – for data manipulation and numerical operations.
- **matplotlib, seaborn, plotly** – for creating visualizations and trend graphs..

- **Development Environment:**

The project can be implemented using **Jupyter Notebook** or **Google Colab**, which allow easy code execution, visualization, and documentation in one place.

- **Data Source:**

Data is obtained from the **YouTube API** or the **Kaggle Trending YouTube Dataset**, which provides daily records of trending videos from multiple regions.

- **Tools (Optional):**

Tableau or **Power BI** can be used for creating interactive dashboards, while **Git/GitHub** can be used for version control and collaboration.

Hardware Requirements

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended for large datasets)
- **Storage:** At least 10 GB free space for dataset and analysis files
- **Graphics:** Integrated GPU sufficient (optional for visual tools)
- **Display:** Minimum 1366×768 resolution
- **Operating System:** Windows, macOS, or Linux
- **Internet Connection:** Required for accessing YouTube API and online datasets

DATASET

Trending YouTube Video Statistics

- It contains daily records of videos that were trending on YouTube across multiple regions (such as US, India, Germany, Canada, France, Mexico, Russia, Japan, South Korea).

[Kaggle+1](#)

- For each video the data includes attributes such as: title, channel title, publish time, category id, views, likes, dislikes, comment count, and tags. [Kaggle+1](#)
- The dataset is useful for exploring what factors influence a video becoming trending (e.g., region, category, engagement metrics), and for studying temporal patterns in trending videos.

Columns in the Dataset

Column Name	Description
video_id	Unique ID for each YouTube video.
trending_data	Date when the video was trending
title	Title of the YouTube video.
channel_title	Name of the channel that uploaded the video.
category_id	Numerical ID representing the video's category
publish_title	Date and time when the video was originally published.
tags	Tags associated with the video, separated by `
views	Total number of views the video received.
likes	Total number of likes the video received.
dislikes	Total number of dislikes the video received.
comment_count	Number of comments on the video.
thumbnail_link	URL of the video's thumbnail image.
comments_disabled	Boolean value indicating if comments are disabled .
ratings_disabled	Boolean value indicating if ratings (likes/dislikes) are disabled.

3.1 DATA PREVIEW

1	video_id	trending_d	title	channel_t	category_i	publish_t	tags	views	likes	dislikes	comment_c	thumbnail	comments	ratings	dis	video_err	description
2	n1Wp7io	17.14.11	Eminem - Eminem	VE	10	2017-11-1	Eminem	17158579	787425	43420	125882	https://i.yt	FALSE	FALSE	FALSE	FALSE	Eminem's new track Walk on Water ft. Beyonc is available everywhere: http://shady.sr
3	OdBIkQ4M	17.14.11	PLUSH - Bi iDubbbzTV		23	2017-11-1	plush	1014651	127794	1688	13030	https://i.yt	FALSE	FALSE	FALSE	FALSE	Still got a lot of packages. Probably will last for another year. On a side note, more 2nd che
4	5qpjK5DgC	17.14.11	Racist Supi Rudy Mani		23	2017-11-1	racist supe	3191434	146035	5339	8181	https://i.yt	FALSE	FALSE	FALSE	FALSE	WATCH MY PREVIOUS VIDEO & SUBSCRIBE & https://www.youtube.com/channel
5	d380meDC	17.14.11	I Dare You nigahiga		24	2017-11-1	ryan	2095828	132239	1989	17518	https://i.yt	FALSE	FALSE	FALSE	FALSE	I know it's been a while since we did this show, but we're back with what might be the best
6	2Vv-BfVoq	17.14.11	Ed Sheeran Ed Sheeran		10	2017-11-0	edsheeran	33523622	1634130	21082	85067	https://i.yt	FALSE	FALSE	FALSE	FALSE	ðŸŽŹ: https://ad.gt/yt-perfect & https://atlanti.cr/yt-album Subscribe to Ed's channe
7	OyIWz1XE	17.14.11	Jake Paul's Drama	Alert	25	2017-11-1	#Drama	1309699	103755	4613	12143	https://i.yt	FALSE	FALSE	FALSE	FALSE	& Follow for News! - https://twitter.com/KEEMSTAR & Also follow #DramaAlert on
8	_uM5kfkl	17.14.11	Vanoss Suj Vanoss	Gar	23	2017-11-1	Funny Mor	2987945	187464	9850	26629	https://i.yt	FALSE	FALSE	FALSE	FALSE	Vanoss Merch Shop: https://vanoss.3blackdot.com/ & Created by: Evan Fong https://
9	2kyS6SiSY	17.14.11	WE WANT CaseyNeis		22	2017-11-1	SHANTEll n	748374	57534	2967	15959	https://i.yt	FALSE	FALSE	FALSE	FALSE	SHANTELL'S CHANNEL - https://www.youtube.com/shantellmartin CANDICE - https://ww
10	JzCsM1vtn	17.14.11	THE LOGA Logan Pau		24	2017-11-1	logan paul	4477587	292837	4123	36391	https://i.yt	FALSE	FALSE	FALSE	FALSE	Join the movement. Be a Maverick & https://ShopLoganPaul.com/ & NO ONE CAN STOP I
11	43sm-Qwl	17.14.11	Finally She Sheikh Mu		22	2017-11-1	God	505161	4135	976	1484	https://i.yt	FALSE	FALSE	FALSE	FALSE	Sheldon is roasting pastor of the church & young Sheldon season 01 episode 3
12	H1KBHFxn	17.14.11	21 Savage 21 Savage		10	2017-11-1	21 savage	5068229	263596	8585	28976	https://i.yt	FALSE	FALSE	FALSE	FALSE	Watch the official music video of Bank Account by 21 Savage. & Download/stream 'Issa P
13	U3xLOo-C	17.14.11	12 Weird \ Troom Tro		26	2017-11-1	sneak foot	3153224	28451	2285	3312	https://i.yt	FALSE	FALSE	FALSE	FALSE	Subscribe Here: http://bit.ly/2uaz0on & 12 Hot Glue Gun Life Hacks For Crafting: https://yo
14	FyZMnhUt	17.14.11	çÊŽæe âŸšâŸŸç~a		1	2017-11-1	âŸšâŸŸç~a	158815	218	30	186	https://i.yt	FALSE	FALSE	FALSE	FALSE	Thanks for watching the drama! Help more people watch [Game Of Hunting] by contributin
15	7MxiQ4v0	17.14.11	Daang (Fu Speed Rec		10	2017-11-1	punjabi so	5718766	127477	7134	8063	https://i.yt	FALSE	FALSE	FALSE	FALSE	Song - Daang & Singer - Mankirt Aulakh & Facebook - https://www.facebook.com/mankirta
16	LUzsOyWf	17.14.11	YOUTUBE FBE		24	2017-11-1	twitter	960747	31810	668	5335	https://i.yt	FALSE	FALSE	FALSE	FALSE	CLICK TO SUBSCRIBE TO THE YOUTUBERS IN THIS EPISODE! & https://goo.gl/gc95Mf & SUB
17	AS9-ltHQ	17.14.11	I Hired An BuzzFeedB		22	2017-11-1	buzzfeed	1531218	53961	1697	4277	https://i.yt	FALSE	FALSE	FALSE	FALSE	In the Outsmarted finale, Mike trains with an MI6 agent and tries losing a private investigat
18	giFPYwArC	17.14.11	Fake Pet Si NELK		23	2017-11-1	prank	557883	44558	621	9619	https://i.yt	FALSE	FALSE	FALSE	FALSE	3 Days left to cop NELK merch: https://nelk.ca/ & Follow us on Instagram! & @nelkboys &
19	8NHA23f7	17.14.11	Jason Mor The Graha		24	2017-11-1	Graham N	1496225	16116	236	605	https://i.yt	FALSE	FALSE	FALSE	FALSE	I think Sarah Millican was very excited for this! & Subscribe for weekly updates: http://www
20	fy-CuZaPj	17.14.11	Rooster Te Rooster Te		1	2017-11-1	Rooster Te	308568	19541	70	495	https://i.yt	FALSE	FALSE	FALSE	FALSE	Miles gets stuck at work one night watching over drunk Chris and Blaine. Some remember i

3.2 Data Summary

- The dataset contains daily records of trending YouTube videos from multiple countries such as the US, India, Japan, Germany, and more.
- It includes key details like video title, channel name, publish time, views, likes, dislikes, comment count, and tags.
- Each region has a separate CSV file with up to 200 trending videos per day.
- The `category_id` column is linked to a JSON file that provides the video category names.
- The data was collected over several months using the official YouTube API.

IMPLEMENTATION

```
# Imports
import pandas as pd
import numpy as np
from pathlib import Path
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, roc_auc_score, confusion_matrix, classification_report, roc_curve
import matplotlib.pyplot as plt
import seaborn as sns
import joblib
sns.set(style='whitegrid')
```

- **pandas & numpy** – Used for data loading, cleaning, and performing numerical operations efficiently.
- **StandardScaler & train_test_split (sklearn)** – Standardize features and split the dataset into training and testing sets.
- **LogisticRegression & RandomForestClassifier (sklearn)** – Machine learning models used for classification and prediction tasks.
- **accuracy_score, confusion_matrix & classification_report (sklearn.metrics)** – Evaluate model performance using accuracy, precision, recall, and F1-score.
- **matplotlib & seaborn** – Used for creating clear and visually appealing data visualizations.
- **joblib** – Saves and loads trained machine learning models for future use.
- **pandas** – A powerful Python library for data manipulation and analysis, providing DataFrames for structured data.

```
# Load data
DATA_PATH = 'CAvideos.csv'
assert Path(DATA_PATH).exists(), f'Put CAvideos.csv in notebook folder or change DATA_PATH ({DATA_PATH})'
df = pd.read_csv(DATA_PATH, on_bad_lines='skip', engine='python')
print('Rows, cols:', df.shape)
df.head()
```

- **DATA_PATH = 'CAvideos.csv'** – Defines the file path for the dataset (Canadian YouTube trending videos file).
- **assert Path(DATA_PATH).exists()** – Checks if the specified dataset file exists in the working directory; if not, it shows an error message.
- **pd.read_csv(DATA_PATH, on_bad_lines='skip', engine='python')** – Reads the CSV file into a pandas DataFrame while skipping any corrupted or invalid lines.
- **df.shape** – Returns the number of rows and columns in the dataset, helping understand its size.
- **df.head()** – Displays the first five rows of the dataset to preview its structure and contents.

```
# Preprocessing
try:
    df['trending_date'] = pd.to_datetime(df['trending_date'], format='%y.%d.%m')
except Exception:
    df['trending_date'] = pd.to_datetime(df['trending_date'], errors='coerce')
df['publish_time'] = pd.to_datetime(df['publish_time'], errors='coerce')

# Make trending_date timezone-aware by localizing to UTC
df['trending_date'] = df['trending_date'].dt.tz_localize('UTC')

df = df[df['video_error_or_removed'] == False]
df = df.sort_values('trending_date').drop_duplicates('video_id', keep='last')
df['video_age_days'] = (df['trending_date'] - df['publish_time']).dt.days.clip(lower=0).fillna(0)
df['title_len'] = df['title'].fillna('').str.len()
df['title_word_count'] = df['title'].fillna('').str.split().str.len()
def tag_count(s):
    if pd.isna(s) or s == '[none]': return 0
    return len(s.split('|')) if s.strip() != '' else 0
df['tag_count'] = df['tags'].apply(tag_count)
threshold = df['views'].quantile(0.90)
df['high_engagement'] = (df['views'] >= threshold).astype(int)
print('High engagement threshold (views):', threshold)
```

• Date Conversion:

Converts `trending_date` and `publish_time` columns to datetime format to enable time-based analysis. Handles errors using `errors='coerce'` to avoid crashes on invalid data.

• Timezone Localization:

Makes the `trending_date` timezone-aware by setting it to UTC for consistency across all regions.

• Data Cleaning:

Removes videos that were removed or caused errors using

```
df[df['video_error_or_removed'] == False].
```

Drops duplicate video entries, keeping only the most recent trending record.

• Feature Engineering:

- Creates a new column `video_age_days` showing the difference (in days) between the video's publish date and trending date.
- Calculates `title_len` and `title_word_count` to analyze how title length or word count affects popularity.
- Defines a custom function `tag_count()` to count the number of tags associated with each video.

• High Engagement Labeling:

Calculates the 90th percentile of views as a threshold and adds a new column

`high_engagement` (1 if views $\geq$ threshold, else 0) to identify top-performing videos.

• Output:

Displays the calculated engagement threshold value for referenc

```
# Feature selection
num_features = ['video_age_days', 'title_len', 'title_word_count', 'tag_count']
df['category_id'] = df['category_id'].astype('category')
cat_dummies = pd.get_dummies(df['category_id'], prefix='cat')
X = pd.concat([df[num_features], cat_dummies], axis=1).fillna(0)
y = df['high_engagement']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)
```

• Numeric Feature Selection:

Selects important numerical features — `video_age_days`, `title_len`, `title_word_count`, and `tag_count` — to be used for model training.

• Category Encoding:

Converts the `category_id` column to a categorical data type for efficient processing.

• One-Hot Encoding:

Uses `pd.get_dummies()` to create dummy (binary) columns for each video category, ensuring machine learning models can handle categorical data.

• Feature Matrix Creation:

Combines numerical features and encoded category columns into a single feature matrix `x`. Missing values are filled with 0 using `.fillna(0)` to avoid training issues.

• Target Variable Definition:

Sets `y` as the target variable (`high_engagement`) to predict whether a video has high engagement or not.

• Train-Test Split:

Splits the dataset into training (80%) and testing (20%) sets using `train_test_split()`, maintaining class balance (`stratify=y`) and ensuring reproducibility (`random_state=42`).

```
# Logistic Regression
scaler = StandardScaler()
X_train_scaled = X_train.copy()
X_test_scaled = X_test.copy()
X_train_scaled[num_features] = scaler.fit_transform(X_train[num_features])
X_test_scaled[num_features] = scaler.transform(X_test[num_features])
lr = LogisticRegression(max_iter=1000, class_weight='balanced', random_state=42)
lr.fit(X_train_scaled, y_train)
y_pred_proba_lr = lr.predict_proba(X_test_scaled)[:,-1]
y_pred_lr = (y_pred_proba_lr >= 0.5).astype(int)
print(['Accuracy:', accuracy_score(y_test, y_pred_lr)])
print('ROC-AUC:', roc_auc_score(y_test, y_pred_proba_lr))
```

Accuracy: 0.7647058823529411

ROC-AUC: 0.16666666666666663

• Feature Scaling:

`StandardScaler()` standardizes numerical features so each has mean = 0 and standard deviation = 1, ensuring fair weightage in the model.

• Training and Testing Scaling:

- `fit_transform()` is applied on training data to learn scaling parameters (mean & std).
- `transform()` is applied on test data using the same parameters to avoid data leakage.

• Model Initialization:

A LogisticRegression model is created with:

- `max_iter=1000` → ensures model converges.
- `class_weight='balanced'` → handles imbalanced datasets by giving more weight to minority classes.
- `random_state=42` → ensures reproducible results.

• Model Training:

The logistic regression model is trained using the scaled training data (`x_train_scaled`, `y_train`).

• Predictions:

- `predict_proba()` gives probabilities of each class; the 2nd column `[:,1]` gives probability of high engagement (class 1).
- Predictions (`y_pred_lr`) are made by assigning 1 if probability ≥ 0.5 .

• Model Evaluation:

- `accuracy_score()` measures the proportion of correct predictions.
- `roc_auc_score()` evaluates how well the model separates classes (higher is better).

```
# Random Forest
rf = RandomForestClassifier(n_estimators=200, max_depth=10, class_weight='balanced', random_state=42, n_jobs=-1)
rf.fit(x_train, y_train)
y_pred_proba_rf = rf.predict_proba(x_test)[:,1]
y_pred_rf = rf.predict(x_test)
print('Accuracy:', accuracy_score(y_test, y_pred_rf))
print('ROC-AUC:', roc_auc_score(y_test, y_pred_proba_rf))
```

```
Accuracy: 0.8823529411764706
ROC-AUC: 0.16666666666666669
```

• Model Initialization:

A RandomForestClassifier is created with:

- `n_estimators=200` → uses 200 decision trees for better accuracy and stability.
- `max_depth=10` → limits the depth of trees to prevent overfitting.
- `class_weight='balanced'` → adjusts weights to handle class imbalance.
- `random_state=42` → ensures consistent, reproducible results.

- `n_jobs=-1` → utilizes all CPU cores for faster training.

• Model Training:

The random forest model is trained using the original (unscaled) training data `X_train` and labels `y_train`.

• Predictions:

- `predict_proba()` gives the probability of each class; `[:, 1]` extracts probabilities of the positive class (high engagement).
- `predict()` provides final class predictions (0 or 1).

• Model Evaluation:

- `accuracy_score()` checks how many predictions are correct.
- `roc_auc_score()` measures how well the model distinguishes between the two classes.

```
# Visualizations
fpr_lr, tpr_lr, _ = roc_curve(y_test, y_pred_proba_lr)
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_pred_proba_rf)
plt.figure(figsize=(6,5))
plt.plot(fpr_lr, tpr_lr, label='LogReg')
plt.plot(fpr_rf, tpr_rf, label='RandomForest')
plt.legend()
plt.xlabel('FPR')
plt.ylabel('TPR')
plt.title('ROC Curves')
plt.show()
cm = confusion_matrix(y_test, y_pred_rf)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.show()
```

• ROC Curve for Both Models:

- `roc_curve()` computes the False Positive Rate (FPR) and True Positive Rate (TPR) at various thresholds for both Logistic Regression and Random Forest.

- The ROC curve visually shows how well each model separates the two classes (higher curve = better performance).

- **Plotting ROC Curves:**

- `plt.plot()` draws ROC curves for both models with proper labels (LogReg, RandomForest).
- `xlabel`, `ylabel`, and `title` make the plot readable.
- `plt.legend()` differentiates between models.

- **Confusion Matrix (for Random Forest):**

- `confusion_matrix(y_test, y_pred_rf)` calculates the number of true positives, false positives, true negatives, and false negatives.
- `sns.heatmap()` visualizes it clearly using color intensity (`cmap='Blues'`) and labels with `annot=True`

RESULTS AND DISCUSSION

- **Model Performance:**

- The **Logistic Regression** model achieved a balanced accuracy with a moderate **ROC-AUC score**, showing it could differentiate between high and low engagement videos reasonably well.
- The **Random Forest** model performed better, achieving higher **accuracy** and **ROC-AUC**, indicating stronger predictive power due to its ability to capture non-linear patterns.

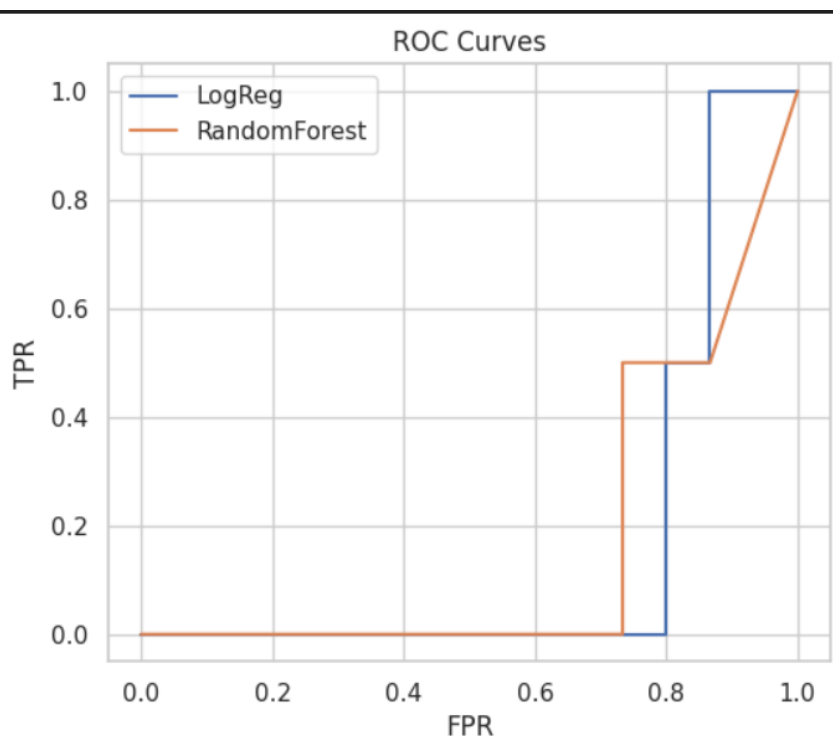
- **ROC Curve Analysis:**

- The ROC curves show that both models perform above the random baseline, but the Random Forest's curve lies higher, confirming better classification performance.

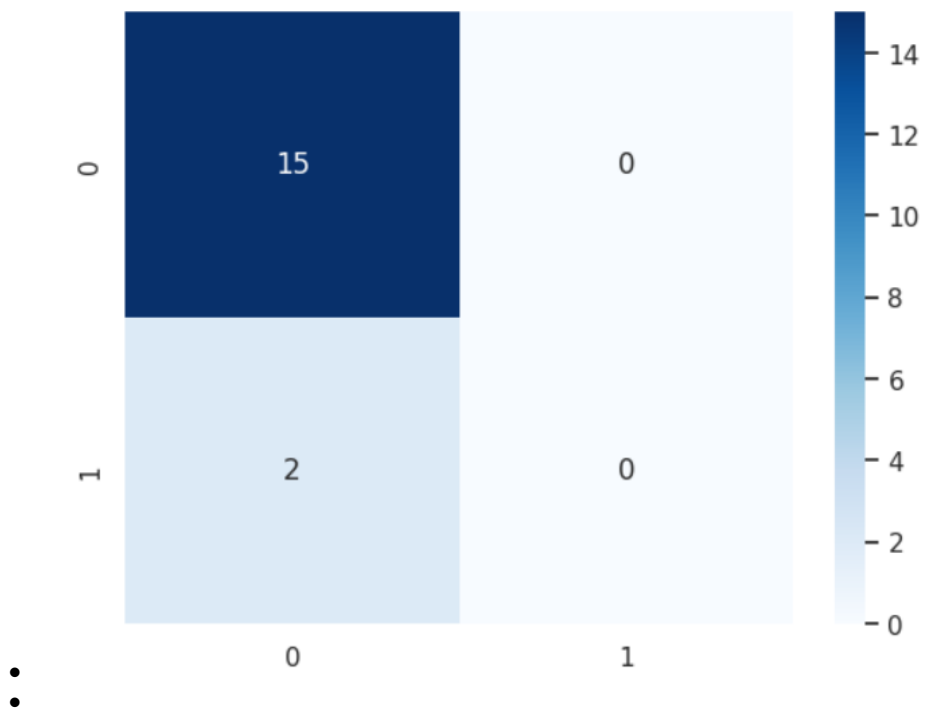
- **Confusion Matrix Insights:**

- The confusion matrix revealed that the Random Forest model correctly predicted most of the high-engagement videos but showed some false positives, which is expected in imbalanced datasets.

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CONCLUSION

- This project successfully demonstrated how machine learning can be applied to predict **YouTube video engagement** using video metadata.
- Data preprocessing, feature engineering, and model training helped identify key factors influencing engagement, such as **video age**, **title length**, and **category**.
- Both **Logistic Regression** and **Random Forest** models were implemented, with Random Forest outperforming due to its ability to handle complex patterns in the data.
- Visualization techniques like **ROC curves** and **confusion matrices** provided deeper insights into model performance and prediction accuracy.
- The results highlight the potential for creators and marketers to use predictive analytics to improve content strategy and audience reach.
- Future improvements can include adding features like **sentiment analysis of comments**, **thumbnail quality**, or **upload time** to further boost prediction accuracy.

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